

Exercise Set 2.1

In each of 1, 2, 3, and 4 represent the common form of each argument using letters to stand for component sentences, and fill in the blanks so that the argument in part (b) has the same logical form as the argument in part (a).

1. a. If all integers are rational, then the number 1 is rational.

All integers are rational.

Therefore, the number 1 is rational.

- b. If all algebraic expressions can be written in prefix notation, then

.

Therefore, $(a + 2b)(a^2 - b)$ can be written in prefix notation.

2. a. If all computer programs contain errors, then this program contains an error.

This program does not contain an error.

Therefore, it is not the case that all computer programs contain errors.

- b. If _____, then _____.

2 is not odd.

Therefore, it is not the case that all prime numbers are odd.

3. a. This number is even or this number is odd.

This number is not even.

Therefore, this number is odd.

- b. _____ or logic is confusing.

My mind is not shot.

Therefore, _____.

4. a. If the program syntax is faulty, then the computer will generate an error message.

If the computer generates an error message, then the program will not run.

Therefore, if the program syntax is faulty, then the program will not run.

- b. If this simple graph _____, then it is complete.

If this graph _____, then any two of its vertices can be joined by a path.

Therefore, if this simple graph has 4 vertices and 6 edges, then _____.

5. Indicate which of the following sentences are statements.

a. 1,024 is the smallest four-digit number that is a perfect square.

b. She is a mathematics major.

c. $128 = 2^6$

d. $x = 2^6$

Write the statements in 6, 7, 8, and 9 in symbolic form using the symbols $\sim$, $\vee$, and $\wedge$ and the indicated letters to represent component statements.

6. Let s = "stocks are increasing" and i = "interest rates are steady."

a. Stocks are increasing but interest rates are steady.

b. Neither are stocks increasing nor are interest rates steady.

7. Juan is a math major but not a computer science major.

(m = "Juan is a math major," c = "Juan is a computer science major")

8. Let h = "John is healthy," w = "John is wealthy," and s = "John is wise."

a. John is healthy and wealthy but not wise.

b. John is not wealthy but he is healthy and wise.

c. John is neither healthy, wealthy, nor wise.

d. John is neither wealthy nor wise, but he is healthy.

e. John is wealthy, but he is not both healthy and wise.

9. Let p = " $x > 5$," q = " $x = 5$," and r = " $10 > x$."

a. $x \geq 5$

b. $10 > x > 5$

c. $10 > x \geq 5$

10. Let p be the statement "DATAENDFLAG is off," q the statement "ERROR equals 0," and r the statement "SUM is less than 1,000." Express the following sentences in symbolic notation.

- a. DATAENDFLAG is off, ERROR equals 0, and SUM is less than 1,000.
- b. DATAENDFLAG is off but ERROR is not equal to 0.
- c. DATAENDFLAG is off; however, ERROR is not 0 or SUM is greater than or equal to 1,000.
- d. DATAENDFLAG is on and ERROR equals 0 but SUM is greater than or equal to 1,000.
- e. Either DATAENDFLAG is on or it is the case that both ERROR equals 0 and SUM is less than 1,000.

11. In the following sentence, is the word *or* used in its inclusive or exclusive sense? A team wins the playoffs if it wins two games in a row or a total of three games.

Write truth tables for the statement forms in 12, 13, 14, and 15.

12. $\sim p \wedge q$

13. $\sim(p \wedge q) \vee (p \vee q)$

14. $p \wedge (q \wedge r)$

15. $p \wedge (\sim q \vee r)$

Determine whether the statement forms in 16, 17, 18, 19, 20, 21, 22, 23, and 24 are logically equivalent. In each case, construct a truth table and include a sentence justifying your answer. Your sentence should show that you understand the meaning of logical equivalence.

16. $p \vee (p \wedge q)$ and p

17. $\sim(p \wedge q)$ and $\sim p \wedge \sim q$

18. $p \vee \mathbf{t}$ and $\mathbf{t}$

19. $p \wedge \mathbf{t}$ and p

20. $p \wedge \mathbf{c}$ and $p \vee \mathbf{c}$

21. $(p \wedge q) \wedge r$ and $p \wedge (q \wedge r)$

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$

23. $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$

24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$

Use De Morgan's laws to write negations for the statements in 25, 26, 27, 28, 29, and 30.

25. Hal is a math major and Hal's sister is a computer science major.

26. Sam is an orange belt and Kate is a red belt.

27. The connector is loose or the machine is unplugged.

28. The train is late or my watch is fast.

29. This computer program has a logical error in the first ten lines or it is being run with an incomplete data set.

30. The dollar is at an all-time high and the stock market is at a record low.

31. Let s be a string of length 2 with characters from $\{0, 1, 2\}$, and define statements a , b , c , and d as follows:

a = "the first character of s is 0"

b = "the first character of s is 1"

c = "the second character of s is 1"

d = "the second character of s is 2".

Describe the set of all strings for which each of the following is true.

a. $(a \vee b) \wedge (c \vee d)$

b. $(\sim (a \vee b)) \wedge (c \vee d)$

c. $(\sim a)(\vee b) \wedge (c \vee (\sim d))$

Assume x is a particular real number and use De Morgan's laws to write negations for the statements in 32, 33, 34, 35, 36, and 37.

32. $-2 < x < 7$

33. $-10 < x < 2$

34. $x < 2$ or $x > 5$

35. $x \leq -1$ or $x > 1$

36. $1 > x \geq -3$

37. $0 > x \geq -7$

In 38 and 39, imagine that num_orders and $num_instock$ are particular values, such as might occur during execution of a computer program. Write negations for the following statements.

38. $(num_orders > 100 \text{ and } num_instock \leq 500) \text{ or } num_instock < 200$

39. $(num_orders < 50 \text{ and } num_instock > 300) \text{ or } (50 \leq num_orders < 75 \text{ and } num_instock > \dots)$

Use truth tables to establish which of the statement forms in 40, 41, 42, and 43 are tautologies and which are contradictions.

40. $(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$

41. $(p \wedge \sim q) \wedge (\sim p \vee q)$

42. $((\sim p \wedge q) \wedge (q \wedge r)) \wedge \sim q$

43. $(\sim p \vee q) \vee (p \wedge \sim q)$

44. Recall that $a < x < b$ means that $a < x$ and $x < b$. Also $a \leq b$ means that $a < b$ or $a = b$. Find all real numbers that satisfy the following inequalities.

a. $2 < x \leq 0$

b. $1 \leq x < -1$

45. Determine whether the statements in (a) and (b) are logically equivalent.

a. Bob is both a math and computer science major and Ann is a math major, but Ann is not both a math and computer science major.

b. It is not the case that both Bob and Ann are both math and computer science majors, but it is the case that Ann is a math major and Bob is both a math and computer science major.

46. * Let the symbol $\oplus$ denote *exclusive or*; so $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$. Hence the truth table for $p \oplus q$ is as follows:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

F

F

F

a. Find simpler statement forms that are logically equivalent to $p \oplus p$ and $(p \oplus p) \oplus p$.

b. Is $(p \oplus q) \oplus r \equiv p \oplus (q \oplus r)$? Justify your answer.

c. Is $(p \oplus q) \wedge r \equiv (p \wedge r) \oplus (q \wedge r)$? Justify your answer.

47. * In logic and in standard English, a double negative is equivalent to a positive. There is one fairly common English usage in which a “double positive” is equivalent to a negative. What is it? Can you think of others?

In 48 and 49 below, a logical equivalence is derived from [Theorem 2.1.1](#). Supply a reason for each step.

$$\begin{aligned}
 48. \quad & (p \wedge \sim q) \vee (p \wedge q) \equiv p \wedge (\sim q \vee q) \quad \text{by (a)} \\
 & \equiv p \wedge (q \vee \sim q) \quad \text{by (b)} \\
 & \equiv p \wedge \mathbf{t} \quad \text{by (c)} \\
 & \equiv p \quad \text{by (d)}
 \end{aligned}$$

Therefore, $(p \wedge \sim q) \vee (p \wedge q) \equiv p$.

$$\begin{aligned}
 49. \quad & (p \vee \sim q) \wedge (\sim p \vee \sim q) \\
 & \equiv (\sim q \vee p) \wedge (\sim q \vee \sim p) \quad \text{by (a)} \\
 & \equiv \sim q \vee (p \wedge \sim p) \quad \text{by (b)} \\
 & \equiv \sim q \vee \mathbf{c} \quad \text{by (c)} \\
 & \equiv \sim q \quad \text{by (d)}
 \end{aligned}$$

Therefore, $(p \vee \sim q) \wedge (\sim p \vee \sim q) \equiv \sim q$.

Use [Theorem 2.1.1](#) to verify the logical equivalences in 50, 51, 52, 53, and 54. Supply a reason for each step.

$$50. (p \wedge \sim q) \vee p \equiv p$$

$$51. p \wedge (\sim q \vee p) \equiv p$$

$$52. \sim(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$$

$$53. \sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$$

$$54. (p \wedge (\sim(\sim p \vee q))) \vee (p \wedge q) \equiv p$$

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